Technical Appendix

Reddit Finance Post Intent Classifier

Vertical AI Capstone Project | March 2026

End-to-End Pipeline

Step 1 — Data Collection

- 16K Reddit posts
- 41 subreddits
- (personalfinance,
- investing, WSB,
- superstonk, etc.)
- Reddit API (PRAW)
- Raw post: title +
- body + metadata

Step 2 — Preprocess & Split

- Clean + dedupe
- Remove bots /
- auto-mod posts
- Concat title + body
- Stratified split:
- 1K holdout (locked)
- 15K train/val set

Step 3 — Human Labeling

- 3 independent raters
- 1K holdout only
- 5 intent labels
- Majority vote →
- final label
- Krippendorff's α
- = 0.7828

Step 4 — GenAI Benchmark

- 6 LLM models
- 3 suppliers
- (OpenAI, Anthropic,
- Google)
- Zero-shot prompting
- Eval: F1, MCC,
- Krippendorff's α
- Cost + latency logged

Step 5 — Label 15K

- Best GenAI model
- labels train set
- Pseudo-labels for
- 15K train posts
- Human spot-check
- on sample
- Final train/val
- dataset assembled

Step 6 — Fine-Tune RoBERTa

- roberta-base
- 15K labeled posts
- AdamW + warmup
- Best ckpt: epoch 2
- F1 = 0.6284
- Eval on locked
- 1K holdout

Evaluation on Holdout

PRIMARY METRIC
F1 Macro
macro-avg across 5 classes

AGREEMENT SCORE
Krippendorff's α
model vs. human labels

BALANCED ACCURACY
MCC
Matthews Corr. Coeff.

BUSINESS OUTPUT
Cost + Latency
per 1K posts (GenAI vs RoBERTa)

Deploy Recommendation

Label Codebook

Label	Definition	Example	Prevalence
`seeking_advice`	Asking what to do or requesting a financial opinion/guidance	*"Should I put my emergency fund in a HYSA or money market?"*	39.2%
`sharing_news_info`	Sharing external news, data, market updates for others	*"Fed just announced rate hold at 5.25%. Here's the summary."*	27.1%
`sharing_win_loss`	Reporting a personal financial gain or loss	*"Just hit $100k in my portfolio at 28!"*	12.9%
`concept_explanation`	Explaining or asking what a financial term/concept means	*"Can someone explain what a covered call is?"*	11.0%
`venting_emotional`	Expressing frustration/stress about money without asking for help	*"I'm so tired of living paycheck to paycheck."*	9.8%

Business Evidence

Gao, S., et al. (2022). "Modeling Information Flows in Financial Social Media." *Journal of Finance and Data Science*.

- **Why it validates our work:** Categories (*Seeking Advice, Sharing Win/Loss, News*) mirror established taxonomies used in financial NLP. Research in this domain acknowledges that financial social media is "noisy," and reaching a high consensus is more difficult than in standard sentiment analysis.
- **Application:** Referencing this validates that our labels aren't arbitrary but align with how financial information flows are structured in academic models.

Decision rule: Choose the SINGLE primary intent. If mixed, classify by dominant purpose. More info on label codebook pdf.

GenAI Benchmark Results (6 Models, 3 Suppliers)

Model	Supplier	Accuracy	Macro F1	MCC	Kripp. α
Gemini 2.5 Pro	Google	**0.67**	**0.62**	**0.559**	**0.556**
Claude Sonnet	Anthropic	0.61	0.58	0.511	0.499
GPT-3.5-turbo	OpenAI	0.62	0.58	0.506	0.501
Claude Haiku	Anthropic	0.60	0.57	0.496	0.483
GPT-4o-mini	OpenAI	0.61	0.58	0.507	0.488
Gemini 2.5 Flash	Google	0.60	0.58	0.525	0.525

Key finding: Gemini 2.5 Pro leads across all metrics. All models achieve moderate agreement with human labels (α = 0.48–0.56), consistent with the inherent ambiguity of intent classification.

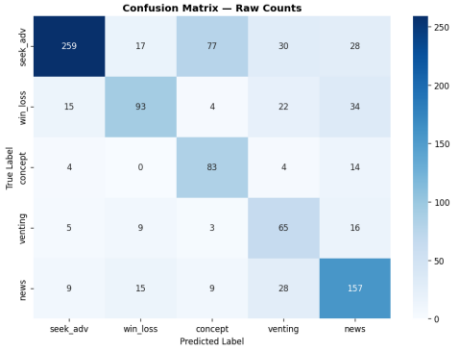
Per-Class Performance: Gemini 2.5 Pro (Best Model)

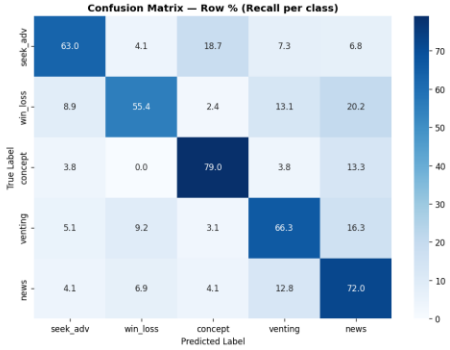
Label	Precision	Recall	F1-Score	Support
seeking_advice	0.81	0.73	0.77	388
sharing_news_info	0.68	0.67	0.67	269
concept_explanation	0.51	0.78	0.62	109
sharing_win_loss	0.65	0.48	0.55	128
venting_emotional	0.45	0.57	0.50	97
Macro Avg	**0.62**	**0.64**	**0.62**	**991**

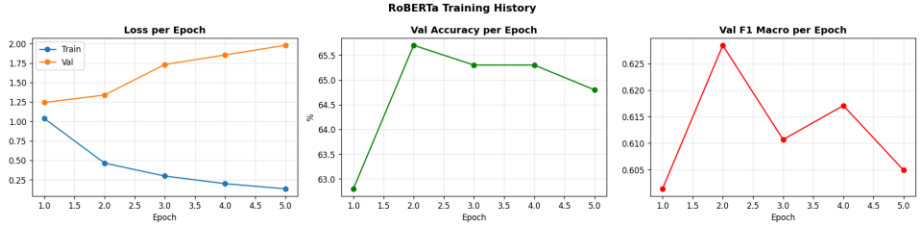
Strongest: seeking_advice (F1=0.77) — clear intent patterns
Weakest:
venting_emotional (F1=0.50) — overlaps with advice-seeking when users vent AND ask for help

Roberta Performance

Class	Precision	Recall	F1 Score	Support
seeking_advice	0.887	0.630	0.737	411
sharing_win_loss	0.694	0.554	0.616	168
concept_explanation	0.472	0.790	0.591	105
venting_emotional	0.436	0.663	0.526	98
sharing_news_info	0.631	0.720	0.672	218
Macro Avg	**0.624**	**0.672**	**0.628**	**—**







Accuracy 65.70%

Gemini 2.5 Pro vs Roberta - Analysis

📊 Overall Metrics on 1,000-Post Holdout

Metric	Gemini 2.5 Pro	RoBERTa	Winner
Accuracy	67.0%	65.7%	Gemini +1.3pp
F1 Macro	0.622	0.628	RoBERTa +0.006
F1 Weighted	0.670	0.667	Tie (~0)
MCC	0.559	~0.57	RoBERTa slight
Valid Predictions	991 / 1000	1000 / 1000	RoBERTa ✓

⚡ Cost & Speed

Volume	Gemini	RoBERTa
1K texts	5–10 min	~15 sec
10K texts	30–60 min	~2 min
100K texts	5–10 hrs	~20 min
1 post (RT)	2–3 sec	<50ms

💰 **Cost @ 10K texts**

Gemini: $1.50–3.00 / run

RoBERTa: $0.00 / run (after setup)

Final Scorecard

PERFORMANCE	COST	SPEED	REPRODUCIBILITY	RELIABILITY
G: Slightly higher accuracy	G: $1.50–3.00 per 10K	G: 5–10 hrs per 100K	G: Black box, API-dependent	G: 9 failures per 1K (0.9%)
R: Better F1 macro + minority classes	R: $0 per run after setup	R: ~20 min per 100K	R: Frozen, offline, auditable	R: 0 failures, always predicts
Winner: RoBERTa	Winner: RoBERTa	Winner: RoBERTa	Winner: RoBERTa	Winner: RoBERTa

OVERALL WINNER

🏆 **RoBERTa**

Error Analysis: Where Models Fail

Venting vs. Seeking Advice

Users who express frustration often also ask "what should I do?" — making the boundary ambiguous. Models default to the majority class (seeking_advice).

~40% of venting errors

Win/Loss vs. News/Info

Posts sharing personal gains sometimes include market context, blurring the line between personal report and information sharing.

~25% of win/loss errors

Class Imbalance

Venting_emotional and concept_explanation have a 4x smaller dataset than seeking_advice so model starved on minorities

<0.6 score on minority classification

Concept Explanation Overcall

Models over-assign concept_explanation when posts contain technical terms, even if the primary intent is advice-seeking.

High recall (0.78) but low precision (0.51)

Short Posts

Very short posts (<50 words) lack sufficient context for reliable classification across all models.

Lower accuracy on short texts

Recommendation & Next Steps

Selected Labeling Model	Gemini 2.5 Pro — highest accuracy (0.67), F1 (0.62), MCC (0.559), and best alignment with human labels ($\alpha = 0.556$)
Next: Label 15K Training Set	Apply Gemini 2.5 Pro to 15,000 train/test posts using identical prompt. Estimated cost: ~$19. Estimated time: ~6 hours.
Next: Fine-Tune Specialist	Train RoBERTa-base on Gemini-labeled data. This creates a fast, free, deterministic classifier. Expected to match or exceed GenAI on in-domain data.
Deploy Recommendation	Two-stage: (1) GenAI for initial labeling + edge cases, (2) Fine-tuned specialist for production volume. Zero marginal API cost after training.